U.S. Capital Markets

Profit-Yield Regime Monitor

Methodology, Implementation, and Interpretation

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Abstract

The Profit-Yield Regime Monitor studies the interaction between U.S. corporate profit growth and long-term Treasury yields. It uses public data from the U.S. Bureau of Economic Analysis and the Federal Reserve Bank of St. Louis to classify the market environment into one of four regimes. The project is built as a small, reproducible research exercise. The data sources, signal definitions, classification rule, and limitations are stated explicitly.

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1 Research Question

The monitor addresses a simple question:

> *What does the joint direction of corporate profit growth and long-term government bond yields imply about the current market regime?*

Rising profits are usually associated with stronger economic activity and support for equities. Rising long-term yields can signal stronger nominal growth, higher inflation risk, tighter financial conditions, or a higher term premium. The two signals should therefore be interpreted together.

The project does not forecast security returns. It organizes two observable signals into a weekly market note.

2 Why Use a Public Earnings Proxy

2.1 Target concept

Analyst earnings revisions are useful because they measure changes in forward-looking expectations. High-quality historical estimate data are usually supplied by commercial vendors. A public GitHub project should not assume access to proprietary data.

2.2 Selected proxy

The monitor uses Corporate Profits After Tax from the Bureau of Economic Analysis, available through FRED as series `CP`. This series is:

- public;
- quarterly;
- seasonally adjusted at an annual rate;
- available from 1947; and
- revised as national accounts are updated.

The proxy measures realized, economy-wide profits. It is not equivalent to forward-looking S&P 500 analyst estimates. The distinction is shown directly in the application and README.

2.3 Trade-off

	Analyst estimates	**BEA proxy**
Orientation	Forward-looking	Realized
Coverage	Listed index companies	U.S. corporations
Frequency	Frequent revisions	Quarterly
History	Vendor-dependent	Public history from 1947
Revisions	Forecast changes	National-account revisions
Use in this project	Target concept	Implemented signal

3 Data

3.1 Series

Series	**Description**	**Frequency**
`CP`	Corporate profits after tax	Quarterly
`DGS10`	10-year Treasury yield	Daily
`DGS2`	2-year Treasury yield	Daily
`T10YIE`	10-year breakeven inflation	Daily

Table 2: Public FRED series used by the monitor

The app downloads each series from FRED's public CSV endpoint. It does not require an API key.

3.2 Frequency alignment

Treasury and breakeven data are converted to weekly Friday observations using the last available value. Quarterly profit observations are carried forward until a new observation becomes available. This creates a weekly table while preserving the quarterly frequency of the profit signal.

The resulting weekly date does not mean profit data were updated that week. It means the latest available quarterly profit observation was used for that week's classification.

4 Signal Construction

4.1 Profit growth

Year-over-year corporate profit growth is:

$$g_t = 100 \left(\frac{CP_t}{CP_{t-4}} - 1 \right).$$

Profit acceleration is the change in year-over-year growth:

$$a_t = g_t - g_{t-1}.$$

The profit signal is:

$$P_t = \begin{cases} \text{accelerating}, & a_t > 0, \\ \text{decelerating}, & a_t \leq 0. \end{cases}$$

This signal measures whether the growth rate of profits is improving, not whether the level of profits is positive.

4.2 Yield trend

Let y_t^{10} be the 10-year Treasury yield and let L be the selected lookback in weeks. The change is:

$$\Delta y_t^{10} = 100 \left(y_t^{10} - y_{t-L}^{10} \right)$$

in basis points. The yield signal is:

$$Y_t = \begin{cases} \text{rising}, & \Delta y_t^{10} > 0, \\ \text{falling}, & \Delta y_t^{10} \leq 0. \end{cases}$$

The default lookback is 13 weeks. The app also offers 4-, 8-, and 26-week windows.

4.3 Supporting indicators

The 2s10s curve slope is:

$$\text{Slope}_t = 100 \left(y_t^{10} - y_t^{2} \right)$$

in basis points. A positive value indicates a conventional upward-sloping curve.

The 10-year breakeven rate is displayed as a market measure of average inflation compensation over the next decade. Neither supporting indicator changes the regime classification in version 0.1.

5 Four-Regime Framework

Profits	10Y yield	Regime	Interpretation
Accelerating	Rising	Structural growth / higher cost of capital	Nominal growth is stronger, but discount rates and the cost of capital are also rising.
Accelerating	Falling	Risk-on easing	Profit momentum is improving while financial conditions become more supportive.
Decelerating	Rising	Margin pressure / stagflation risk	Profit momentum is weakening while discount rates remain under upward pressure.
Decelerating	Falling	Cyclical slowdown	Both profit momentum and long-term yields point toward slower nominal growth.

The classification rule is intentionally deterministic:

```
def classify_regime(profits_accelerating, yields_rising):
    if profits_accelerating and yields_rising:
        return "Structural growth / higher cost of capital"
    if profits_accelerating:
        return "Risk-on easing"
    if yields_rising:
        return "Margin pressure / stagflation risk"
    return "Cyclical slowdown"
```

6 Processing Pipeline

The calculation runs in the following order:

1. Download the four FRED CSV files.
2. Parse dates and convert values to numeric observations.
3. Calculate quarterly year-over-year profit growth.
4. Calculate the quarterly change in profit growth.
5. Resample the daily rate series to weekly Friday observations.
6. Carry the latest quarterly profit signal into the weekly table.
7. Calculate the lookback change in the 10-year yield.

8. Calculate the 2s10s slope.
9. Classify every week into a regime.
10. Display the current regime and historical series.

7 Application Design

7.1 Interface

The Streamlit application contains:

- a sidebar for chart history and yield lookback;
- a current-regime card with a short interpretation;
- five metric cards;
- a dual-axis historical chart;
- colored background bands showing past regimes;
- a table of recent weekly observations;
- a CSV download; and
- an expandable methodology note.

7.2 Chart

The chart places year-over-year corporate profit growth on the left axis and the 10-year Treasury yield on the right axis. Background color indicates the classified regime.

The two axes have different units and should not be compared by vertical distance. The chart is intended to show direction, turning points, and regime transitions.

7.3 Caching

Downloaded data are cached for one hour:

```
@st.cache_data(ttl=3600, show_spinner=False)
def get_history(lookback_weeks):
    data = load_market_data()
    return build_history(...)
```

Caching reduces repeated network requests when a user changes a display control.

8 Repository Structure

```
profit-yield-regime-monitor/
|-- app.py
|-- regime_monitor/
|   |-- __init__.py
|   |-- data.py
|   |-- event_study.py
|   +-- signals.py
|-- tests/
|   +-- test_signals.py
|-- .streamlit/config.toml
|-- .github/workflows/tests.yml
|-- requirements.txt
|-- pyproject.toml
|-- README.md
+-- documentation.tex
```

`data.py` handles downloads, while `signals.py` handles the classification. The event calculations live in `event_study.py`. Presentation stays in `app.py`. This separation allows the analytical logic to be tested without running Streamlit or making a network request.

9 Running the Project

```
git clone <repository-url>
cd profit-yield-regime-monitor
python -m venv .venv

# macOS or Linux
source .venv/bin/activate

# Windows PowerShell
.venv\Scripts\Activate.ps1

pip install -r requirements.txt
streamlit run app.py
```

Streamlit normally opens at http://localhost:8501.

10 Testing

Run:

```
python -m unittest discover -s tests -v
```

The tests cover:

Four regime combinations
Confirms that each pair of Boolean signals maps to the expected regime.

Signal calculations
Uses synthetic quarterly profits and daily rates to check positive yield change, curve slope, and the resulting regime.

Empty input
Confirms that requesting a snapshot from an empty history raises an error.

Event dates and summaries
Checks that only regime-entry dates become events and that the grouped sample counts agree with the event-level data.

GitHub Actions runs the same tests on every push and pull request.

11 Interpreting the Output

11.1 Structural growth / higher cost of capital

This regime does not mean that higher yields are harmless for equities. It means the profit and yield signals may share a common source, such as stronger nominal activity, supply constraints, or investment demand. Equity outcomes still depend on whether earnings growth is sufficient to offset higher discount rates.

11.2 Risk-on easing

Improving profit momentum and falling yields are jointly supportive. The classification does not distinguish falling yields caused by lower inflation from those caused by concern about future growth. Breakevens and the yield curve provide additional context.

11.3 Margin pressure / stagflation risk

This is the least favorable combination in this classification. Slower profit growth reduces fundamental support while rising yields increase discount rates. The label describes a signal combination, not a formal stagflation forecast.

11.4 Cyclical slowdown

Falling yields can support valuations, but decelerating profits point to weaker activity. The net asset-price effect is not predetermined.

12 Event Study

The dashboard includes a short event study to check what followed each regime in the available ETF history. An event occurs only when the regime label changes. This avoids treating every week in a long regime as an independent observation.

For asset j, event date t, and a 13-week horizon, the forward price return is

$$R_{j,t}^{(13)} = 100 \left(\frac{P_{j,t+13}}{P_{j,t}} - 1 \right).$$

The two assets are SPY, used as a U.S. equity proxy, and TLT, used as a long-duration Treasury proxy. Daily adjusted closes are downloaded from Yahoo Finance and resampled to Friday observations. For each regime and asset, the app reports the number of events, median return, mean return, standard deviation, and share of positive returns.

The chart uses the median because a few large market moves can pull the mean away from a typical event. The event count is shown with the result; small groups should not be read as stable estimates.

This exercise is descriptive. It does not include transaction costs and relies on the latest vintage of the revised profit series. Regimes may also overlap in their 13-week outcome windows when signals change quickly. The results are a historical check on the classification, not evidence of a trading strategy.

13 Limitations

1. **The profit proxy is backward-looking.** BEA profits do not measure analyst revisions.
2. **Mixed frequencies create stale signals.** The profit signal changes quarterly while yields change daily.
3. **Data are revised.** Historical classifications may change after BEA revisions.
4. **The thresholds are binary.** A one-basis-point yield change and a large yield move receive the same direction label.
5. **The lookback is selected by the user.** Different windows can produce different regimes.
6. **No asset-return forecast is estimated.** The labels are market descriptions, not trading signals.
7. **The model is U.S.-focused.** Global profit, currency, and policy differences are not included.
8. **The event study uses ETF adjusted closes.** The sample begins with available ETF data and depends on a third-party price source.
9. **Event windows can overlap.** Closely spaced regime changes do not produce statistically independent observations.

14 Possible Extensions

The project can remain compact while adding one extension at a time:

- add neutral zones around zero to reduce frequent switching;
- report regime persistence and transition counts;
- compare 4-, 13-, and 26-week event horizons;
- add a real-yield series;
- compare BEA profits with a user-supplied analyst-estimate CSV; or
- record data vintages to study the effect of revisions.

15 Data Sources

- U.S. Bureau of Economic Analysis, Corporate Profits: BEA corporate profits page
- FRED Corporate Profits After Tax: FRED series CP
- FRED 10-Year Treasury Yield: FRED series DGS10
- FRED 2-Year Treasury Yield: FRED series DGS2
- FRED 10-Year Breakeven Inflation: FRED series T10YIE
- Yahoo Finance Historical Prices: Yahoo Finance

16 Conclusion

The monitor is deliberately narrow. It takes one question from market commentary—how profits and yields are moving relative to one another—and turns it into two explicit signals. The result is easy to reproduce and easy to disagree with. Every judgment is visible, from the earnings proxy to the 13-week yield window.

Disclaimer
For research only. Data may be revised. This project does not provide investment advice.